

IoT Devices
(Sensors, Cameras, Thermostats,
Appliances)

Raw Data Input

Neural Perception Module
(CNNs, RNNs, LSTMs, VAEs)

Processed Features

Parameter Refinement

Integration Interface
(Neural to Symbolic Transformation,
Feedback Loop for Refinement)

Symbolic Representations

Feedback for Refinement

Symbolic Reasoning Module
(Knowledge Base, Rule-Based
Inference Engine, Logical
Constraints)

Decisions/Actions

Decision Output
(Actions to IoT Devices or
Environment)

Control/Actuation

